FUTURE IMPLEMENTATION / 01

What comes after the first build

Ranked by feasibility of the bounded version described, easiest first. These are engineering judgments, not time estimates, market-novelty claims or permission to expand iteration one.

Baseline already implemented

The first build has a profile/text evidence workspace, a six-call demo catalog plus manually added calls, explainable matching, a cited checklist with document statuses, an agenda/.ics export, three draft sections and deterministic readiness checks. Everything below extends that baseline. Higher production scale or scientific ambition can change the ranking.

Recommended sequence

01 Full proposal templates and reusable content - high feasibility
02 Document vault and manual ethics records - high
03 Resubmission response matrix - high
04 Evidence-linked AI rubric review - high for a bounded assistive version
05 Award obligations and progress reports - moderate
06 Budget planning and expenditure ledger - moderate
07 Scheduled reminders - moderate; deployment dependent
08 Sponsor financial-report preparation - moderate; ledger dependent
09 Live opportunity discovery - moderate to difficult
10 Team workspaces and institutional connectors - difficult
11 GrantWeave: compatible funding combinations - difficult
12 AimFit: study design under funding limits - difficult; domain dependent
13 Bootstrap: cheapest valid preliminary-evidence plan - most demanding

How to use the order

Choose one bounded release at a time. The first four primarily reuse existing text, versions and tasks. Numbers 5-8 introduce award data, money and reliable operations. Numbers 9-10 add changing external systems and access control. The last three require explicit constraint or scientific models. AimFit and GrantWeave can swap order if validated formulas and domain expertise make one easier. The roadmap is a dependency-aware default, not a mandate to ship every ordinary feature before testing a novel concept.

Common release gate

For each feature, demonstrate one complete workflow with persisted state, source traceability and visible uncertainty. Exercise normal, missing-data, stale-source and failure cases. Separate deterministic calculations from model suggestions. Do not call a demo feed live, a plan an experiment, or an exported report certified. Measure user time saved and corrections required; do not infer award-rate improvement from a demo or an AI score.

FUTURE IMPLEMENTATION / 02

Complete proposals and documents

The first expansion should deepen the existing workflow before introducing new systems.

01 / Full proposal templates and reusable content

Outcome: prepare the complete narrative packet for a supported sponsor rather than only three starter sections.
Inputs: a researcher-confirmed call/template, section order and limits, approved project facts, reusable bios/facilities text and prior accepted draft versions.
Bounded build: support one or two explicit sponsor templates. Add section-level requirements, reusable content with provenance, controlled updates across applications, and DOCX/PDF export through a validated layout. Preserve a human review step when reusing text. Do not automatically claim that a previous project's results or approvals apply to a new project.
Output: a versioned proposal packet, unresolved-input list and clear mapping of each section to the call. Existing evidence and readiness checks apply to every section.
Dependencies: initial draft/version model, call requirements and a reliable export renderer.
Acceptance: section order, limits and headings match the selected template; reused content remains traceable; a source edit marks affected sections stale; an export preserves headings and tables without losing unresolved-input warnings.
Limit: supporting a specific template does not establish support for every sponsor form, attachment package or electronic submission portal.

02 / Document vault and manual ethics records

Outcome: connect required attachments and factual claims to specific file versions, and see missing or expired supporting records.
Inputs: PDF/TXT/DOCX files, category, project/application, owner, version/date, plus manually confirmed approval scope, status and expiry where relevant.
Bounded build: allowlisted, size-limited uploads; text extraction; searchable metadata; requirement-to-file links; revision history. Track IRB, animal-study, data-use, training and conflict-of-interest records as user-confirmed statuses. Add OCR only after normal parsing is reliable, with visible uncertainty and review of extracted metadata.
Output: a document inventory, supported claim links, missing/stale attachment notices and approval-expiry tasks.
Dependencies: protected file storage, safe parsing, version records and access checks before any hosted use.
Acceptance: replacing a file preserves referenced old versions; an unreadable file is retained but excluded from drafting evidence; an expired record displays its scope/date without inventing a project-wide prohibition; only a human confirms an approval.
Limit: tracking ethics records cannot issue approval, determine legal compliance or replace the institution's process. Formal submission/approval integrations belong to item 10. Sensitive study data requires an appropriate storage and privacy design.

FUTURE IMPLEMENTATION / 03

Revise and review applications

Extend the existing draft and source model. Keep administrative completeness separate from scientific critique.

03 / Resubmission response matrix

Outcome: convert a rejected application into a traceable revision plan.
Inputs: the exact submitted proposal, reviewer comments/summary statement, target call and current project facts.
Bounded build: paste or upload comments; propose distinct concerns with exact source quotes; let the researcher accept tasks, owners, affected sections and evidence needs. Link revisions to concerns and export an editable response document. Explicitly distinguish a formal resubmission from a new application to another program.
Output: concern -> proposed action -> supporting evidence -> revised section -> response, with unresolved status maintained until the researcher records completion.
Dependencies: proposal snapshots, source references, checklist tasks and drafting. Add file parsing only through item 02 if needed; pasted text is enough for an initial version.
Acceptance: every extracted concern resolves to the original comment; cosmetic rewriting does not close a missing-evidence concern; a response never claims a new experiment was completed without a recorded result; old and new versions remain comparable.
Limit: addressing all comments cannot guarantee funding. Suggested revisions may conflict with each other or the new call; surface the conflict for a decision.

04 / Evidence-linked AI rubric review

Outcome: identify unclear claims, inconsistent facts and weak support before submission.
Inputs: a frozen draft, the actual call rubric, selected scientific context and evidence records.
Bounded build: one structured review pass plus existing deterministic checks. Return criterion-level findings, draft spans, rationale, severity, uncertainty and suggested actions. Prefer statuses and findings. If adding a numeric grade, display the explicit rubric, label it a tool-generated assessment and retain prior runs; never label it an award probability.
Output: a review report with accepted/dismissed findings and a revision task list. Separate administrative blockers from AI scientific suggestions.
Dependencies: stable draft snapshots, confirmed call criteria and evidence retrieval. Add a domain-expert evaluation set before making scientific-quality claims.
Acceptance: test planted contradictory sample sizes, an unsupported pilot result and a missing section; flag relevant passages and admit when evidence is insufficient. A rerun must not silently overwrite accepted findings.
Limit: a plausible model critique is not calibrated peer review. Better wording, a higher model grade or a passed checklist does not establish scientific validity or a likely panel score.

FUTURE IMPLEMENTATION / 04

Manage the funded project

Awarded terms, actual progress and actual spending are new facts. Keep them separate from what the proposal requested.

05 / Award obligations and progress-update drafting

Outcome: convert a notice of award into manageable obligations and evidence-backed progress reports.
Inputs: award agreement, approved scope/budget, reporting template, milestone plan and recorded research updates.
Bounded build: one award per application; proposed obligations with source references and human confirmation; editable milestone statuses; one progress-report template. Keep proposed, approved and amended scope/budget versions separate. Use the existing agenda before requiring external notifications.
Output: reporting schedule, responsible person, obligations checklist and an editable narrative describing completed, delayed and unreported work.
Dependencies: source/document versions, tasks, project facts and drafting. Text-only award intake can precede a full vault.
Acceptance: importing an award does not overwrite the proposed budget; delayed milestones are described honestly; unsupported results remain missing inputs; the accepted report retains its underlying evidence snapshot.
Limit: a proposal does not prove work was performed. Sponsor submission, certification and acceptance remain explicit human actions. Start with narrative reporting; financial reporting is item 08.

06 / Budget planning and expenditure ledger

Outcome: understand requested, awarded, spent, committed and remaining amounts by category and period.
Inputs: proposed budget, approved award and amendments, category/rate assumptions, expenses, commitments and a defined transaction CSV format.
Bounded build: one currency per award; explicit periods and category mappings; exact decimal or integer-minor-unit money; deterministic personnel/equipment/other-cost totals. Separate actual expenses from commitments and forecasts. Include optional receipt links, duplicate import detection and an audit trail. User confirms indirect-cost base and rates rather than AI guessing them.
Output: proposal budget, award budget-versus-actual ledger, remaining balance and explanatory warnings. Budget revisions never rewrite transaction history.
Dependencies: award records, deterministic calculation layer and versioning. Reconciliation to an institutional source is required before reporting authoritative balances.
Acceptance: repeated imports do not double-count; rejected rows are visible; refunds/credits work; totals reconcile exactly; negative balances remain visible; amendments retain before/after values; an expense is not counted again as a commitment.
Limit: this is a project ledger, not an institutional accounting system. A positive remaining balance does not establish that a cost is allowable or funds are legally available.

FUTURE IMPLEMENTATION / 05

Reliable reminders and financial reports

Both features look simple in a demo. Their value depends on reliable state and explicit operational checks.

07 / Scheduled reminders outside the app

Outcome: receive a reminder without keeping the application open.
Inputs: confirmed deadlines, chosen lead times, timezone, explicit recipient/channel and opt-in preferences.
Bounded build: a single channel such as email, one periodic scheduler, persistent notification outbox, idempotency keys, bounded retries, unsubscribe/pause controls and visible delivery state. Distinguish provider acceptance from confirmed delivery where the provider cannot prove delivery.
Output: scheduled reminders, delivery/failure history and recoverable retry actions.
Dependencies: deployed worker/scheduler, durable storage, identity and configured delivery provider. Existing .ics and in-app agenda remain usable before this is deployed.
Acceptance: repeating a scheduler run does not duplicate a reminder; changing a deadline updates/cancels pending reminders; failures are visible; timezone and daylight-saving boundary tests preserve timing; withdrawing opt-in prevents future sends.
Limit: browser timers are not reliable background jobs. Reminders cannot be promised before scheduling and provider credentials are configured. Calendar OAuth and bidirectional synchronization are separate connector work under item 10.

08 / Sponsor financial-report preparation

Outcome: prepare a reconciled financial draft in one funder's reporting format.
Inputs: approved budget amendments, reconciled transactions, reporting period, sponsor category mappings and authorized adjustments.
Bounded build: one explicit CSV/XLSX or document template. Compute all numbers in deterministic code; use AI only to explain verified variances. Define the accounting basis and period cutoff. Keep reconciliation status, review approval and the immutable ledger snapshot used for each report.
Output: a draft statement/report, category totals, variance narrative and traceable adjustments, marked for finance review.
Dependencies: item 06's ledger, budget versioning and exports; access to actual sponsor requirements and institutional review.
Acceptance: totals reconcile to the selected ledger snapshot; no transaction disappears during category mapping; period changes recalculate correctly; unsupported adjustments are excluded or flagged; later transactions do not silently change a saved report.
Limit: a generated budget-versus-actual statement is not an audited financial statement, every sponsor's official form or a GAAP reporting system. Authorized finance staff must review and certify reports. Submission automation is a later, separate integration.

FUTURE IMPLEMENTATION / 06

Fresh opportunities and shared work

External data and multiple users introduce ongoing operational work, not just more interface screens.

09 / Live grant discovery and feed synchronization

Outcome: find new and amended opportunities without manually maintaining the catalog.
Inputs: one documented official feed/API, researcher profiles, source metadata and ingestion configuration.
Bounded build: first replace demo-only usefulness with a small manually verified real catalog. Then add one source-specific API adapter with caching, deduplication, versions and scheduled refresh. Verify actual API access and usage terms at implementation time. Separate a discovery record from authoritative call text; detect changed deadlines and closed notices. Reuse the existing tri-state eligibility and explainable relevance flow.
Output: new/changed opportunities, source-linked match reasons, freshness indicators and amendment alerts for saved applications.
Dependencies: initial matching, ingestion jobs, adapter and any required credentials; reliable parsing for supported notices.
Acceptance: repeated ingestion is idempotent; an amendment updates the same opportunity while preserving old versions; outages do not relabel cached data as current; contradictory source dates require review; an excellent topic match cannot override a known eligibility mismatch.
Limit: a single feed does not cover all foundations or agencies. Broad scraping, comprehensive coverage and continuously correct eligibility are separate data-quality challenges. Do not equate semantic similarity with eligibility.

10 / Team workspaces and institutional connectors

Outcome: collaborate with co-investigators/research offices and reduce duplicate entry into existing systems.
Inputs: authenticated users/roles, workspace memberships, audit policy and explicit access to one selected external system.
Bounded build: ship private owner/editor/viewer workspaces with server-enforced permissions, comments, review requests and conflict-safe edits first. Then add one read-only connector with a declared source of truth. Treat document-drive sync, accounting sync, calendar OAuth and IRB-system integration as separate releases. Never assume an institution exposes a public API.
Output: permission-aware shared applications, change history and controlled synchronization with visible conflicts/errors.
Dependencies: migration from local single-user storage, multi-tenant authorization, credential/token management, audit records and actual institutional access.
Acceptance: changing record IDs cannot expose another workspace; revoked membership blocks future access; retries do not duplicate imported records; stale external updates cannot overwrite newer accepted changes; sync failures preserve local work.
Limit: integration access, institutional contracts and privacy requirements may dominate engineering effort. Formal ethics approval still comes from the authorized institution, even if its status is synchronized.

11 / GrantWeave

Construct a conditional funding combination for a researcher-confirmed plan. High potential value; difficult because budget arithmetic is only part of compatibility.

The decision it supports

Can these specific opportunities jointly support this research plan, and which restrictions prevent it? Explain both feasible allocations and unsatisfied constraints. Do not present the sum of grant maxima as money the researcher has secured.

Inputs and outputs

Inputs: a small set of work packages with scientific scope, cost categories, dates, personnel effort and existing support; three to five calls/awards; source-linked, researcher-confirmed restrictions; dependencies between work packages.
Outputs: alternative allocations, unfunded gaps, date/effort conflicts, assumptions and an explanation of each binding restriction. Separate already awarded funds from hypothetical award scenarios. A failed funding scenario should show which parts of the plan stop or need replacement funding.

Smallest credible implementation

Define a typed constraint schema and an exact deterministic solver over a bounded instance. Model allowed categories, award caps, dates, effort capacity, mutually exclusive costs and user-confirmed overlap rules. Require confirmation before parsed terms become constraints. Let the researcher compare a few scenarios and revise assumptions. Explain why no solution exists where possible. No automated grant applications or fund transfers. The kit's analytics allocation utilities can be evaluated for a simple subset; do not assume they already implement this domain model.

Dependencies and acceptance

Needs mature call/award/budget schemas, source provenance and explicit work packages. A live grant feed is optional. Review restrictions with a knowledgeable grants administrator.
Tests: reject ample total funding with incompatible dates; prevent the same expense being allocated twice; enforce total personnel effort; distinguish scientific overlap from merely different expense labels; show a known feasible alternative when one conflicting restriction is removed. Compare solver results with hand-checked small cases and retain an assumptions snapshot.

Limits and later expansion

Feasible means feasible under the encoded, confirmed assumptions. Arithmetic cannot certify that scientific overlap or multiple-award support is permitted. Relabeling the same aim or cost does not resolve prohibited duplication. Add larger portfolios, uncertainty and sequencing only after bounded cases are reliable; never turn an LLM's interpretation of terms directly into a compliance guarantee.

Success measure

A researcher or grants officer can understand the proposed allocation, identify the controlling restrictions and correct an assumption without rebuilding the plan. Measure verified constraint satisfaction and explanation usefulness before claiming improved funding outcomes.

NOVEL PLANNING / 08

12 / AimFit

Compare scientifically defensible study options within a grant ceiling. The first version must support one validated design family.

The decision it supports

What study can this grant actually support, and what conclusions would each affordable design justify? Preserve the research question and make changes to scope explicit. A cheaper plan is useful only if it still addresses a meaningful question.

Inputs and outputs

Inputs: one supported design family, a researcher-defined question, effect/precision assumptions with provenance, fixed and per-unit costs, attrition/resource assumptions, grant cap, timeline and user-approved design alternatives.
Outputs: a small comparison of costed designs, sample size or analogous resource choice, sensitivity ranges, unmet targets and the corresponding scope/claim changes. Expose assumptions so a domain expert can inspect them. Show infeasible results instead of manufacturing an affordable answer.

Smallest credible implementation

Select one established design family with a statistics adviser, such as a specified two-group comparison under explicit assumptions. Implement validated numerical calculations in code, compare with independent reference cases, and enumerate a bounded set of acceptable design variants. AI may explain the tradeoffs and draft proposal edits; it does not invent effect sizes or perform authoritative arithmetic. Require the researcher to approve any changed design before it updates the application.

Dependencies and acceptance

Needs a reliable cost model, evidence provenance, versioned study assumptions, proposal revisions and domain/statistical validation. Full live discovery and institutional integration are not prerequisites.
Tests: match known reference calculations; vary effect, precision, variance and cost assumptions; report when no design meets the target; never preserve an unattainable precision claim after reducing the budget; preserve all original assumptions and the accepted alternative. Have a domain expert assess whether the proposed comparisons remain meaningful.

Limits and later expansion

Statistical power and affordability do not establish scientific merit, validity or practical feasibility. Never optimize repeatedly on observed outcomes to manufacture significance. Recruiting, ethics, instrument access and time can be binding even when the budget works. Wet-lab experiments, clinical trials and arbitrary study redesign require separate validated models; do not claim general support from one calculator.

Success measure

Researchers can explain why a selected design fits the cap, which assumptions matter, and which claims it can and cannot support. Evaluate numerical correctness and expert acceptance before testing whether the tool improves real proposal decisions.

13 / Bootstrap

Find the lowest-cost credible route to missing preliminary evidence. This is the most demanding idea because relevance and scientific validity dominate the optimization.

The decision it supports

What is the next affordable piece of evidence worth obtaining for this proposal? Connect reviewer concerns and evidence gaps to candidate actions, resources, results and later application revisions. Optimize for useful evidence, not favorable findings.

Inputs and outputs

Inputs: a target application, explicit concerns/questions, existing evidence, expert-approved candidate experiments or analyses, estimated costs/durations, available data/equipment and possible seed-funding scenarios.
Outputs: a ranked plan with costs, dependencies, uncertainty and expected relevance to each gap; resource/funding prerequisites; and an evidence record updated only after real work is performed. Distinguish minimum cost within the supplied candidate set from a claim to have found the globally cheapest valid study.

Smallest credible implementation

Choose one computational research workflow and a small reviewed catalog of possible analyses. Start as a planning tool on known inputs: rank actions that may address confirmed gaps, show uncertainty, and let the researcher reject weak links. Later, add execution of one predefined analysis on suitable public data in an isolated environment with versioned code, inputs, logs and outputs. Do not build arbitrary autonomous experimentation. Retain negative and inconclusive results and update the plan honestly.

Dependencies and acceptance

Needs evidence/version tracking, review or resubmission concerns, cost planning and domain validation. GrantWeave can supply funding scenarios and AimFit can supply design alternatives, but neither needs to be general-purpose first.
Tests: choose actions that address declared gaps within confirmed resources; expose uncertain gap-to-test links; include fixed setup costs and failed funding branches; retain failed/inconclusive analyses; draft pilot claims only from actual outputs; recompute consistently after a cost change or rejected seed grant. Have experts assess whether the proposed evidence would be relevant before relying on solver performance.

Limits and later expansion

A perfect optimizer over an incorrect scientific model still gives a bad plan. Preliminary findings may fail to support the hypothesis or may not be persuasive to a funder. Cost estimates need sources, dates and ranges. Never promise favorable results or an award. Expand into new analysis families only with reproducibility checks and independent scientific evaluation.

Success measure

Measure whether experts agree the proposed next step is relevant, feasible and worth its cost; whether execution is reproducible; and whether generated claims faithfully describe results. A useful negative finding counts as evidence, not a product failure.

Build on the same foundations

A connected product depends on consistent facts and versions across the lifecycle, not on every capability shipping at once.

Dependency map in plain language

Proposal templates, resubmissions and rubric review extend the original draft/evidence model. Award reports add approved terms and actual progress. The ledger supplies numbers for financial reports and the novel funding tools. Reminders need deployment infrastructure, while live discovery needs a source adapter. Shared workspaces require an authorization/storage upgrade before institutional integrations. GrantWeave needs confirmed constraints; AimFit needs validated design/cost models; Bootstrap needs reliable gap-to-evidence relationships. Build only the dependencies needed for the next chosen release.

Kit shortcuts to inspect at each stage

documents + extraction + retrieval: file normalization, exact evidence and bounded corpus search.
exports: report/CSV generation; confirm template fidelity and formula-safe spreadsheet export.
integrations + jobs: adapters and bounded background work; application scheduling, recipients and persistence still need implementation.
research + recommendations: safe discovery/extraction primitives and transparent ranking; an official funding-feed adapter remains domain work.
analytics: deterministic calculations and bounded allocation; verify whether the optimizer supports the constraints required by GrantWeave.
supabase: candidate building blocks for authentication/private storage; an example notes schema is not a finished research-workspace authorization design.
ai: structured generation with explicit fixture/live modes; validate evidence and domain claims separately.

Release and evaluation policy

Ship one complete bounded addition, integrate it with existing state, and verify prior workflows still work. Track time to a source-checkable application plan, requirement extraction precision/recall on reviewed calls, unsupported draft claims, date/reminder correctness, financial reconciliation and expert acceptance of planning suggestions. Use synthetic or authorized data in demos. Award-rate improvement requires longitudinal evaluation with appropriate comparisons, not an AI rubric score.

Source and interpretation

Technical reuse is based on the inspected Hackathon Kit catalog and selected module documentation at revision 22a5f44a27375e88ac2cd1754f6976704f725bc5. The feasibility order and proposed product designs are planning judgments. The three named novel functions are product concepts; this document does not claim that no comparable product or research system exists.

Reference

https://github.com/oliverchennn/hackathon-kit
See CATALOG.md, docs/start-here/AGENT_START.md, docs/start-here/CREATE.md and selected modules/<id>/README.md and module.json. Recheck the current repository when implementing; reuse does not remove domain-specific validation, integration or operational work.